

COM-SU-5315-2.0

Three-Layer Field Joint Coating

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1 Scope

This specification defines the general characteristics and minimum technical and performance requirements for the selection and application of anticorrosion coating(s) for pipeline field joints that are compatible with a project's chosen three-layer pipe coating system(s) and intended for subsea service.

1.1 Conflict Resolution and Order of Precedence

1. Conflicts between documents shall be submitted to Owner in writing for resolution. If requirements between different documents are in conflict, in general, the more stringent requirements shall apply, with Owner's concurrence.
2. Deviations and exceptions from codes, standards, and references listed herein shall require Owner notification before they will be considered acceptable.
3. This specification shall comply with all applicable federal, state, and local codes and regulations on surface preparation, coating application, storage, handling, safety, and environmental requirements.
4. In the event of a conflict between this specification and other project documents, the priority for choosing the governing document shall be as follows:
 - a. Contract conditions.
 - b. Scope of work (SOW).
 - c. This specification.
 - d. Referenced drawings.
 - e. Referenced codes and standards.

2 References

2.1 Owner Documents

None

2.2 Industry Codes and Standards

Unless otherwise specified herein, use the latest edition of the referenced documents.

ASTM International (ASTM)

ASTM D512 Standard Test Method for Chloride Ion in Water

ASTM D570 Standard Test Method for Water Absorption of Plastic

ASTM D638 Standard Test Method for Tensile Properties of Plastics

ASTM D792 Standard Test Methods for Density and Specific Gravity (Relative Density) of Plastics by

Displacement

ASTM D1238 Standard Test Method for Melt Flow Rates of Thermoplastics by Extrusion Plastometer

ASTM D2240 Standard Test Method for Rubber Property-Durometer Hardness

ASTM D4060 Standard Test Method for Abrasion Resistance of Organic Coatings by the Taber Abraser

ASTM D4285 Standard Test Method for Indicating Oil or Water in Compressed Air

ASTM D4417 Standard Test Methods for Field Measurement of Surface Profile of Blast Cleaned Steel

ASTM D4940 Standard Test Method for Conductimetric Analysis of Water Soluble Ionic Contamination of Blasting Cleaning Abrasives

ASTM E337 Standard Test Method for Measuring Humidity with a Psychrometer (the Measurement of Wet- and Dry- Bulb Temperatures)

Deutsches Institut Für Normung E.V. (DIN)

DIN 30670 Polyethylene coatings on steel pipes and fittings – Requirements and testing

International Organization for Standardization (ISO)

ISO 179-1 Plastics - Determination of Charpy Impact Properties

ISO 306 Plastics Thermoplastic Materials - Determination of Vicat Softening Temperature (VST)

ISO 868 Plastics and Ebonite – Determination of Indentation Hardness by Means of a Durometer (Shore Hardness)

ISO 1133 Plastics Determination of the Melt Mass-Flow Rate (MFR) and the Melt Volume-Flow Rate (MVR) of Thermoplastics

ISO 8501-1 Preparation of Steel Substrates Before Application of Paints and Related Products—Visual Assessment of Surface Cleanliness;

Part 1—Rust Grades and Preparation Grades of Uncoated Steel Substrates and of Steel Substrates After Overall Removal of Previous Coatings

ISO 8501-3 Preparation of Steel Substrates Before Application of Paints and Related Products—Visual Assessment of Surface Cleanliness;

Part 3—Preparation grades of welds, edges and other areas with surface imperfections

ISO 8502-3 Preparation of Steel Substrates before Application of Paints and Related Products—Tests for the Assessment of Surface Cleanliness;

Part 3—Assessment of Dust on Steel Surfaces Prepared for Painting (Pressure-Sensitive Tape Method)

ISO 8502-6 Preparation of Steel Substrates Before Application of Paints and Related Products—Tests for the Assessment of Surface Cleanliness;

Part 6—Extraction Methods of Soluble Contaminants for Analysis, The Bresle Method

ISO 8502-9 Preparation of Steel Substrates Before Application of Paints and Related Products—Tests for the Assessment of Surface Cleanliness;

Part 9—Field Method for the Conductometric Determination of Water-Soluble Salts

ISO 8503-5 Preparation of Steel Substrates Before Application of Paints and Related Products—Surface Roughness

Characteristics of Blast-Cleaned Substrates;

Part 5—Replica Tape Method for the Determination of the Surface Profile

ISO 21809-1 External Coatings for Buried or Submerged Pipelines Used in Pipeline Transportation Systems;

Part 1—Polyolefin Coatings (3-layer PE and 3-layer PP)

ISO 21809-3 External Coatings for Buried or Submerged Pipelines Used in Pipeline Transportation Systems;

Part 3—Field Joint Coatings

Association for Materials Protection and Performance (AMPP) (previously NACE International)

NACE SP0274 High-Voltage Electrical Inspection of Pipeline Coatings

NACE SP0394 Application, Performance, and Quality Control of Plant-Applied Single-Layer Fusion-Bonded Epoxy External Pipe Coating

Society for Protective Coatings (SSPC)

SSPC SP 1 Solvent Cleaning (Surface Preparation Specification)

SSPC SP 10 / NACE No.2 Near-White Metal Blast Cleaning

SSPC VIS 1 Guide and Reference Photographs for Steel Surfaces Prepared by Dry Abrasive Blast Cleaning

3 Terminology

3.1 Acronyms

AUT Automated Ultrasonic Testing

CP Cathodic Protection

DSC Differential Scanning Calorimetry

FBE Fusion-Bonded Epoxy

HSS Heat Shrink Sleeve

ITP Inspection and Test Plan/Protocol

MFI Melt Flow Index

MPS Manufacturing Procedure Specification

NCR Nonconformance Report

PE Polyethylene

PP Polypropylene

PPT Pre-Production Test/Trial

PQT Prequalification Test

QC Quality Control

3.2 Definitions

Batch

Quantity of material, that is produced by its manufacturer in one continuous production run using the same or identical raw materials, production conditions and are under the same certificate of analysis. Materials from the same batch are generally packaged, shipped, and stored together and are considered by the manufacturer to have the same or identical physical properties and other characteristics.

Maximum batch quantity is 20 metric tons (22 U.S. tons).

Cutback

Length of pipe left uncoated at each end of the pipe joint to allow two pipes to be welded together.

Holiday

A discontinuity in the protective coating that exposes any unprotected metal surface to the environment.

Production Run

Single continuous operation of Sub-Supplier's production line in which field joints are coated without a change of application parameters or equipment or a scheduled or unscheduled interruption lasting greater than 12 hours.

Project Drawings

Drawings provided to Sub-Supplier by Supplier, or prepared by Sub-Supplier and reviewed and accepted by Supplier, that are intended to depict, define, or convey the field joint configuration, coatings, application procedures, processing equipment, or any other subject related to Sub-Supplier's performance of the work or work product.

Project Manager

Employee of Sub-Supplier that is primarily responsible for interfacing with Supplier and managing every aspect of the work on behalf of Sub-Supplier.

Pup

Short section or length of pipe.

Shift

Continuous working period of no more than 12 hours in which production is maintained without changing the personnel performing the work.

Work

Field joint anticorrosion coating testing and application activities to be performed by Sub-Supplier hereunder.

4 General

4.1 Design

4.1-1 - It shall be the responsibility of Sub-Supplier to complete the following actions:

1. Assess project operating conditions, requirements, and Owner's designated Supplier installation methods and equipment.
2. Propose a field joint coating system design, including configuration, materials, composition, and thickness, that will achieve the mechanical integrity, durability, and anticorrosion requirements defined in this specification and scope of work (SOW) for project line pipe and risers.

4.1-2 - Sub-Supplier shall design and specify the physical configuration, thickness, and materials for the field joint coating system based on the properties of Sub-Supplier's proprietary products and application methods.

4.1-3 - Sub-Supplier's field joint coating system design shall, as a minimum, comply with the following criteria:

1. Bond with the pipe coating to form a single, continuous high-integrity pipeline coating system that is impervious to water and capable of sustaining operating loads without cracking, tearing, or disbonding from the pipe coating.
2. Withstand the expected installation loads (J-lay, S-lay, reeling) and any other adverse conditions imposed by Supplier's pipeline fabrication and construction methods and equipment without damage.
3. Withstand direct exposure to seawater at the design operating temperature and pressure for a period equal to the design life of the pipeline without significant degradation of its physical integrity or anticorrosion performance.
4. Be compatible with the cathodic protection (CP) system.

4.1-4 - The physical properties of the field joint coating system shall be as follows:

1. Meet project requirements for durability and physical integrity over life of the pipeline.
2. Be suitable for use under project service conditions.
3. Be compatible with the project's chosen three-layer pipe coating system(s).
4. Be compatible with the Supplier's pipe-handling and pipelay procedures and equipment.

4.1-5 - The final field joint coating system is subject to Supplier and Owner review and acceptance.

4.2 Additional Requirements

4.2-1 - Performance of the field joint coating system shall be supported by rigorous qualification and production testing.

4.2-2 - Sub-Supplier shall provide precise control of the minimum pipe temperature and temperature distribution used for application of the fusion-bonded epoxy (FBE) pipe coating.

4.2-3 - Sub-Supplier shall provide routine coating of pipe pups that have simulated field joints to permit regular production testing and monitoring of field joint coatings and their application.

4.3 Sub-Supplier Questions

4.3-1 - Drawings, specifications, and other documents relating to the work that are provided to Sub-Supplier shall be checked by Sub-Supplier immediately upon receipt, and Sub-Supplier shall notify Supplier and Owner of any discrepancies or unclear provisions therein.

4.3-2 - In the event that Sub-Supplier has questions regarding a provision or requirement of said documents, it shall be Sub-Supplier's responsibility to do the following:

1. Obtain written clarification of the provision or requirement from Supplier and Owner, which shall be final and binding.
2. Review and resolve conflicts with Supplier and Owner before submission of its bid or initiation or continuation of work.

4.4 Supplier Deviations

4.4-1 - Supplier shall obtain Owner's written authorization to deviate from the requirements of this specification, referenced codes and standards, project drawings, and purchase order agreement.

4.4-2 - The final interpretation of this specification shall rest solely with Owner.

4.5 Technical Representatives and Meetings

4.5-1 - Sub-Supplier shall designate a technical representative who is knowledgeable in every aspect of the proposed anti-corrosion coating, polypropylene (PP), or polyethylene (PE) materials and their application processes.

4.5-2 - Technical representative shall attend all meetings and serve as the primary liaison between Owner, Supplier, and Sub-Supplier.

4.5-3 - Supplier may designate the same individual or another employee as project manager to coordinate coating and insulation activities at Sub-Supplier's facilities.

4.5-4 - Sub-Supplier shall maintain minutes of meetings between Supplier, Owner, and Sub-Supplier representatives and promptly distribute them for Owner acceptance.

4.5-5 - Supplier shall designate a technical representative to attend meetings at Sub-Supplier's facilities and discuss activities that can affect pipe handling, installation, field joint welding and preparation, inspection, field joint coating, and other construction-related activities that are the Supplier's responsibility.

4.5-6 - Supplier's technical representative shall be present during field joint coating application to ensure field joint coating is applied properly in accordance with Supplier's recommendations and procedures.

4.6 Owner and Authority

4.6-1 - Supplier shall provide one or more representatives or inspectors to monitor the work.

4.6-2 - Supplier's inspectors shall be granted access to storage sites, application facilities, and testing and inspection facilities involved in Sub-Supplier's performance of the work.

4.6-3 - Owner may also have inspector(s) who will require the same access and interfaces as Supplier.

4.6-4 - Sub-Supplier shall instruct inspector(s) on the safety of Supplier and Owner personnel for areas the personnel may visit and activities they can witness during the performance of their duties.

4.7 Facilities for Supplier and Owner Inspectors

4.7-1 - Where facilities are under Sub-Supplier's control, Sub-Supplier shall provide on-site office space and other appropriate facilities to Supplier and Owner's inspectors for their use during and throughout Sub-Supplier's coating and production activities.

4.7-2 - At a minimum, these facilities shall include the following:

1. Enclosed, heated and cooled, secure offices for at least two inspectors.
2. Separate meeting room.
3. Restroom(s).
4. Normal office furnishings (desks, bookshelves, tables, etc.).
5. International telephone access and secure, high-speed Internet connections.
6. Office equipment and supplies (photo copier, fax, printers, office supplies, etc.).
7. Convenient access to Sub-Supplier's production facilities.
8. Secure, convenient parking.
9. 24-hour, 7-day/week access to each of the above facilities and services.
10. If applicable, transportation to and from airport/hotel.

4.7-3 - Said facilities shall be provided to Supplier and Owner on a continuous basis, beginning at least two weeks before the start of any project-related production, startup and testing (pre-qualification test [PQT], pre-production testing [PPT], etc.) activities, and continuing until at least two weeks after the completion of production.

5 Material Requirements

5-1 - All coating materials shall be supplied and stored in damage-free, tightly sealed containers with the following information clearly marked on the packaging:

1. Manufacturer's name.
2. Description and grade of material.
3. Number of manufacture/batch number and shelf life.

4. Storage requirements (minimum and maximum temperatures, maximum humidity, etc.).
5. Reference to applicable material safety data sheet(s).

5.1 Storage Conditions

5.1-1 - All materials shall be stored under clean, dry, climate-controlled conditions in accordance with the material manufacturer's recommendations and Sub-Supplier's written testing, storage, and handling procedures.

5.2 Batch Testing

5.2-1 - Each consignment of raw material received by Sub-Supplier shall be checked according to the quality plan.

5.2-2 - Sub-Supplier shall confirm (by testing) that every property listed in Table 6 is in accordance with manufacturer's published values and applicable for use per project-specific requirements listed in the scope of work (SOW).

6 Sub-Supplier Responsibilities

6.1 Design Requirements

6.1-1 - Design requirements for pipe coating and insulation systems shall be as specified by the Owner in the scope of work (SOW).

6.1-2 - The principle characteristics of pipe coating and insulation systems shall be as presented in the scope of work (SOW).

6.2 Proposed System Design

6.2-1 - Sub-Supplier's proposed field joint coating system shall consist of one or more layers of polypropylene (PP), polyethylene (PE), or other polymer-based coating applied over a fusion-bonded epoxy (FBE) or liquid epoxy anticorrosion coating.

6.2-2 - A thin layer of adhesive shall be applied between the anticorrosion coating and polymer-based field joint coating to improve adhesion between the fusion-bonded epoxy (FBE) and polymer-based coatings.

6.2-3 - The field joint anticorrosion system shall consist of the following:

6.2-3.1 - A fusion-bonded epoxy (FBE) anticorrosion coating or a liquid primer applied directly to the pipe surface. The use of a polyurethane-based primer in place of FBE shall not be allowed.

6.2-3.2 - A layer of adhesive applied directly over and bonded to the fusion-bonded epoxy (FBE). This layer of adhesive shall not be required when polypropylene (PP) and polyethylene (PE) heat shrink sleeves that include co-

polymer adhesives are used.

6.2-4 - The field joint anticorrosion system shall consist of one or more layers of the following:

1. Solid injection molded polypropylene (IMPP) or polyurethane.
2. Flame spray polypropylene or polyethylene.
3. Polypropylene or polyethylene shrink sleeve(s) with co-polymer adhesives. The use of shrink sleeves shall not be allowed for reeled pipelay.
4. Other molded-polymer-based material (e.g., polyurethane).

6.2-5 - The use of a co-polymer adhesive shall be required for injection-molded polypropylene (PP)- and polyethylene (PE)-based materials.

6.2-6 - Supplier may propose an alternative field joint coating system for Owner's consideration.

6.2-6.1 - Any proposed field joint coating system shall be subject to a full qualification program in accordance with Owner's requirements.

6.2-6.2 - Owner shall have the final decision regarding acceptance of a proposed field joint coating system.

6.3 Procedures and Documentation

6.3.1 General

6.3.1-1 - As part of its proposal, Sub-Supplier shall provide Supplier and Owner with written procedures and documentation detailing the following:

1. Coating, insulation materials, and system(s).
2. Application equipment and processes.
3. Pipe surface preparation.
4. Pipe insulation chamfer end preparation.
5. Inspection, quality control (QC), testing procedures, and equipment.
6. Operator and QC personnel qualifications and training.
7. Personnel safety procedures, processes, and training.
8. Environmental quality procedures and training.
9. Materials handling and storage procedures.
10. Coating and insulation repair methods.
11. Waste material handling and disposal procedures.
12. Proposed prequalification test (PQT) program.
13. Proposed pre-production test (PPT) program.
14. Proposed inspection and test plan (ITP).
15. Manufacturing procedure specification (MPS).
16. Schedule of qualification programs.

6.3.1-2 - Sub-Supplier shall inform Supplier of all materials used in production of, or incorporated into, the field joint

coating and insulation system(s) that are potentially toxic, reactive, polluting, or subject to special handling and disposal requirements.

6.3.1-3 - Sub-Supplier's documentation shall include the manufacturer's original data sheets, material test results, and other relevant data for the materials used in the manufacture of the proposed field joint coating system to support its stated physical and material properties and verify that the coating system design and materials will satisfy Owner's performance requirements and endurance criteria given in item 4.1-5.

6.3.1-4 - Sub-Supplier shall provide Supplier's inspectors with written records documenting storage and transportation conditions for all materials used by Sub-Supplier in the application of pipe coatings.

6.3.1-5 - Sub-Supplier shall provide written procedure(s) for the documentation, testing, storage, and handling of raw materials used in the application of pipe coatings to Supplier for review and acceptance at least ninety days before the start of production.

6.3.1-6 - Sub-Supplier deliverables shall be summarized as follows:

1. Previous test data on similar systems.
2. Raw material certifications.
3. Pipe receipt inspection and test plan (ITP).
4. Pipe handling and storage plan.
5. Application procedures for field joint coating:
 - a. ITP and manufacturing procedure specification (MPS) for prequalification test (PQT).
 - b. PQT report with test results.
 - c. ITP and MPS for production.
 - d. Repair procedure.
6. Stalk handling and storage plan.
7. Schedule for PQTs.
8. Final as-built documentation, including Owner-requested production records, etc.

6.3.2 Production and Quality Control Documentation

6.3.2-1 - Three months before starting production, Sub-Supplier shall provide Supplier and Owner with manufacturing procedure specification (MPS) and inspection and test plan (ITP) for the pre-qualification test (PQT) and pre-production test (PPT) for each field joint coating system and application process provided by Sub-Supplier, for Supplier and Owner review and acceptance.

6.3.2-2 - Production and quality control (QC) inspection documentation and records (i.e., listing field joints coated, raw materials used, inspection and test results, etc.) shall be gathered and made available to Supplier's and Owner's inspectors on a daily basis.

6.3.2-3 - Full reports covering each week's production and the production to date shall be issued to Supplier and Owner weekly during production.

6.3.2-4 - Supplier, Owner, and their inspectors shall have the right to request and review copies of Sub-Supplier's daily production, inspection, and test records and documentation at any time.

6.3.2-5 - Sub-Supplier shall provide Supplier with a detailed quality control (QC) plan and documented procedures for the application, inspection, and monitoring of application of the field joint coating system.

6.3.2-6 - The results of every production test performed by Sub-Supplier shall be recorded, photographed, and provided to Supplier and Owner inspectors on a daily basis or as agreed to by Supplier and Owner.

6.3.3 Final Report

6.3.3-1 - Within two weeks following the conclusion of the work, Sub-Supplier shall prepare and present a full and comprehensive report containing all quality control (QC)- and production-related documentation related to the work to Supplier and Owner.

6.3.4 Production Meetings

6.3.4-1 - Weekly production meetings between Sub-Supplier, Supplier, and Owner's inspector shall be scheduled on site each week for the purpose of reviewing reports, progress of the work, and discussing the work plan for the following week.

6.3.4-2 - Daily informal meetings to update Supplier's and Owner's inspectors on current activities and to discuss any recent concerns of the inspector's shall be conducted on a daily or once-per-shift basis.

6.3.4-3 - Supplier and Owner's inspectors shall have the right to request an unscheduled meeting with Sub-Supplier's project manager at any time to discuss production, testing, or any other concerns.

6.4 Access to Sub-Supplier Personnel

6.4-1 - Supplier and Owner inspectors shall be given access to all Sub-Supplier personnel involved in the work for the purpose of communicating with personnel and monitoring Sub-Supplier's production and inspection activities. Access shall include, but not be limited to, the following:

1. Managers
2. Engineers
3. Line personnel
4. Operators
5. Quality control (QC) inspectors

6.4-2 - Supplier and Owner inspectors shall interact with personnel in a manner that minimizes any disruption of Sub-Supplier's production activities and ensures their actions do not adversely affect the safe conduct of production activities.

7 Pre-Qualification and Pre-Production Testing

7.1 Testing Program

7.1-1 - Sub-Supplier's proposed field joint coating system shall be subjected to a rigorous testing program before startup of production.

7.1-2 - This testing shall include all applicable provisions of the tables in Section 11 and possibly other requirements, depending upon the properties of the coating systems proposed by Sub-Supplier.

7.2 Responsibilities

7.2-1 - Unless otherwise provided in the scope of work (SOW), this testing shall be performed by Sub-Supplier.

7.2-2 - The test procedures, specimen design, and equipment used shall be subject to review and prior written acceptance of Supplier and Owner.

7.2-3 - Testing-related activities shall be witnessed by Supplier's and Owner's inspectors unless this requirement is waived in writing in advance by Supplier and Owner.

7.2-4 - Any waiver given by Supplier and Owner shall only apply to the single activity and event specified in the waiver.

7.3 Testing Components

7.3-1 - The testing program shall consist of the following two parts:

7.3-1.1 - Pre-qualification tests (PQT) to demonstrate that the proposed field joint coating system design, when properly applied, meets Owner's specified technical and performance requirements, is fit for purpose, and suitable for use under intended service conditions and installation loads.

7.3-1.2 - Pre-production tests (PPT) to demonstrate that the field joint coating system applied using Sub-Supplier's production application and inspection procedures, processes, personnel, equipment will satisfy Owner's specified technical and performance requirements during production and under field conditions and is comparable in performance to field joints produced for the pre-qualification tests (PQT).

7.4 Pre-Qualification Testing

7.4.1 General Requirements

7.4.1-1 - Sub-Supplier's pre-qualification test (PQT) program shall include the following:

7.4.1-1.1 - Tests used to measure application quality and performance of field joint coatings and insulation. In most cases, the tests are similar to ones that will be performed for pipe and field joint coatings during production.

7.4.1-1.2 - Medium term, full-scale simulated service tests for one or more coated field joints under design operating conditions. These tests are intended to confirm physical integrity and thermal performance of the pipe and field joint coating.

7.4.1-2 - Unless this requirement is waived by Owner, separate test series shall be performed for each field joint coating system provided by Sub-Supplier.

7.4.1-3 - The pre-qualification test (PQT) program shall also include or incorporate appropriate procedures and tests for coating repair methods and equipment proposed by Sub-Supplier.

7.4.1-4 - Any changes made to the manufacturing procedure specification (MPS) and inspection and test plan (ITP) used for the pre-qualification test (PQT) by Sub-Supplier after qualification shall, at Owner's sole discretion, be subject to retesting and requalification.

7.4.2 Testing Program

7.4.2-1 - The pre-qualification test (PQT) program shall include the following:

7.4.2-1.1 - Pre-production and production tests for the field joint coatings.

7.4.2-1.2 - Full-scale simulated service testing of coated pipe and field joint test specimens at design operating temperature and pressure.

7.4.2-1.3 - For reeled installation only, the following tests shall be performed:

7.4.2-1.3.1 - Full-scale bending tests to confirm that the coated field joints are suitable for installation using the Supplier's proposed installation methods, procedures, and equipment.

7.4.2-1.3.2 - Straightener and creep fatigue tests to confirm that the coated pipe and field joint can withstand the stresses produced by reeling and straightening the pipe.

7.4.2-1.4 - Appropriate tests (to be determined) for the proposed field joint coating repair methods and equipment.

7.4.2-2 - Additional tests may be required, in the opinion of Owner, to qualify or evaluate new, unusual, or potentially adverse aspects of Sub-Supplier's field joint coating system design.

7.4.2-3 - The pre-qualification tests shall be as detailed in Table 1, Table 2, Table 3, and Table 4.

Note: Table 1 through Table 7 are all located in Section 11.

7.4.3 Number of Tests and Test Specimens

7.4.3-1 - Unless otherwise agreed to by Supplier in writing, the pre-qualification test (PQT) program shall include separate test series for each unique pipe and field joint coating system provided by Sub-Supplier.

7.4.3-2 - The number of field joint specimens prepared and tested shall be sufficient to confirm that each field joint coating system provided by Sub-Supplier will satisfy Owner's performance, integrity, durability, and installation requirements.

7.4.3-3 - Six simulated field joints shall be subjected to the pre-qualification tests (PQT) listed in [Table 1](#) and [Table 2](#) for the fusion-bonded epoxy (FBE) coating, combined FBE and adhesive coatings, and fully coated field joints.

7.4.3-4 - Three coated actual field joints shall be subjected to the full-scale bending tests (or standard reeling trials), point load tests, and simulated service tests outlined in [Table 3](#).

7.4.3-5 - The pre-qualification tests (PQT) performed for each test series shall include any planned or proposed production tests not included in [Table 1](#), [Table 2](#), and [Table 3](#).

7.4.3-6 - The tests shall also include any additional tests deemed necessary by Supplier to evaluate the performance of Sub-Supplier's field joint coating system.

7.4.3-7 - In addition to the tests outlined above, each coated full-scale test specimen, including the complete field joints prepared for coating tests in [Table 2](#), shall be sectioned after testing and visually examined for voids, porosity, disbondment between layers and other internal defects.

7.4.3-8 - If any test field joints prepared for each test series fails any of the prescribed pre-qualification tests (PQT), or if any hidden defects or material failures are discovered when the test specimens are sectioned, then the application process, equipment malfunction, or material defect that caused the failure shall be corrected by Sub-Supplier.

7.4.3-9 - The complete failed test series shall be repeated using newly prepared test field joints until a successful outcome is achieved.

7.4.3-10 - A minimum of five full pipe joints containing a total of nine field joints shall be prepared by Sub-Supplier for pre-qualification test (PQT) of each field joint coating system.

7.4.4 Simulated Field Joints

7.4.4-1 - At least two pipe joints in each test series shall be used by Sub-Supplier to assemble test specimens for the tests listed in [Table 1](#) and [Table 2](#).

7.4.4-2 - Each pipe joint shall include three simulated field joints prepared by removing the pipe coatings at three points spaced at approximately equal intervals on the pipe in a way that matches the geometry and dimensions of an actual field joint.

7.4.4-3 - Sub-Supplier shall use the two pipe joints to prepare six test field joints by coating three simulated field joints

on one test pipe with fusion-bonded epoxy (FBE) only, and three field joints on the second pipe with the complete field joint coating system.

7.4.5 Field Joints

7.4.5-1 - At least three pipe joints in each test series shall be used by Sub-Supplier to assemble three field joints for the full-scale tests listed in Table 3.

7.4.5-2 - Each pipe joint shall include one field joint prepared by removing the pipe coatings at the center of the pipe joint in a way that matches the geometry and dimensions of an actual field joint.

7.4.5-3 - After welding, each test field joint shall be prepared and coated with Sub-Supplier's field joint coating system using the accepted application and inspection processes, procedures, materials, and equipment.

7.4.5-4 - The processes, procedures, materials, and equipment used to prepare test field joints shall be identical to those that will be used by Sub-Supplier in production.

7.4.5-5 - The pipe used to prepare the test field joints shall be project pipe or pipe having the same dimensions (thickness and diameter).

7.4.5-6 - The pipe shall be coated in accordance with project specifications by the Sub-Supplier and using application processes, materials, and equipment identical to those that will be used in production, subject to Owner acceptance.

7.4.6 Test Documentation and Report

7.4.6-1 - Sub-Supplier shall provide Supplier and Owner with a proposed manufacturing procedure specification (MPS) and inspection and test plan (ITP) for the pre-qualification test (PQT) of each pipe coating system, for Owner's review and acceptance, at least three months before start of the PQT.

7.4.6-2 - A full report on the results of the pre-qualification test (PQT) shall be presented to Supplier and Owner for review at least six weeks before the start of production for each system.

7.4.6-3 - Sub-Supplier's production manufacturing procedure specification (MPS) and inspection and test plan (ITP) shall be based upon the MPS and ITP prepared and successfully qualified during the pre-qualification test (PQT).

7.4.7 Acceptance Criteria

7.4.7-1 - Sub-Supplier's proposed field joint coating system shall successfully pass each pre-qualification test (PQT) listed in Table 1 and Table 2, each full-scale bending test, other applicable tests described in Table 3, and any other tests to be qualified by Supplier and Owner for use on the project.

7.4.7-2 - For reeled pipelay, all three combined pipe and field joint test specimens in each full-scale test series shall successfully complete a standard reeling trial without incurring any significant damage (such as cracking and disbondment) to the pipe or field joint coatings for the combined pipe and field joint coating system to be qualified for

use by the project.

7.5 Pre-Production Tests

7.5.1 General Requirements

7.5.1-1 - Pre-production test (PPT) program shall include the preparation and testing of simulated field joints that are coated using actual project pipe along with Sub-Supplier's actual production and inspection equipment, personnel, procedures, and processes.

7.5.1-2 - The number of field joint specimens tested shall be sufficient to demonstrate Sub-Supplier's ability to consistently apply each field joint coating system while satisfying Supplier and Owner's testing, integrity, and performance requirements.

7.5.1-3 - Pre-production test (PPT) program shall include each planned production test for field joint coatings.

Note: Pre-production tests (PPT) are detailed in Table 1, Table 4, and Table 5.

7.5.1-4 - The pipe used to prepare the test field joints shall be actual project pipe coated in accordance with project specifications by the Sub-Supplier using their actual production application processes, materials, and equipment.

7.5.2 Testing Program

7.5.2-1 - Each of the twelve simulated field joints described in Section 7.5.3 and Section 7.5.4 shall be subjected to the applicable parts of the production tests listed in Table 4 and Table 5 for fusion-bonded epoxy (FBE) coating and the complete field joint coating system.

7.5.2-2 - Every coated test field joint shall be sectioned after testing and visually examined for voids, porosity, disbondment between layers, and other internal defects.

7.5.2-3 - If any of the test field joints prepared for each test series fails any prescribed pre-production tests (PPT) (as shown in Table 1, Table 4, and Table 5), or any hidden defects or material failures are discovered when the test specimens are sectioned, then the application process, equipment malfunction, or material defect that caused the failure shall be corrected by Sub-Supplier.

7.5.2-4 - The failed tests shall be repeated using newly prepared test field joints until a successful outcome is achieved.

7.5.3 Number of Tests and Test Specimens

7.5.3-1 - The number of field joint specimens tested shall be sufficient to demonstrate Sub-Supplier's ability to consistently apply each field joint coating system while satisfying Owner's performance, integrity, durability, and installation requirements, in the actual production environment.

7.5.3-2 - A minimum of four pipe joints containing a total of 12 field joints shall be prepared by Sub-Supplier for pre-production tests (PPT) of each field joint coating system.

7.5.3-3 - Six of the test field joints prepared by Sub-Supplier shall be coated only with fusion-bonded epoxy (FBE) and six field joints shall be coated with the complete coating system.

7.5.4 Simulated Field Joints

7.5.4-1 - Pre-production tests (PPT) shall be performed using simulated field joints that are coated using actual project pipe and Sub-Supplier's actual production and inspection equipment, personnel, procedures, and processes.

7.5.4-2 - Each of the four pipe joints in each test series shall include three simulated field joints prepared by removing the coating at three approximately equally spaced points on the pipe in a way that matches the geometry and dimensions of the actual field joint.

7.5.5 Test Documentation and Report

7.5.5-1 - Sub-Supplier shall provide Supplier and Owner with a proposed manufacturing procedure specification (MPS) and inspection and test plan (ITP) for pre-production tests (PPT) of each pipe coating system, based upon the MPS and ITP prepared and qualified during the pre-qualification test (PQT).

7.5.5-2 - The manufacturing procedure specification (MPS) and inspection and test plan (ITP) shall be provided at least three months before the start of the pre-production test (PPT) for Owner's review and acceptance.

7.5.5-3 - A full report on the results of the pre-production test (PPT) shall be presented to Supplier and Owner for review at least six weeks before the start of production for each system.

7.5.6 Acceptance Criteria

7.5.6-1 - Sub-Supplier's proposed field joint coating system shall successfully complete and pass each applicable pre-production test (PPT) listed in Table 1, Table 4, and Table 5 to be accepted as ready for production by Owner.

8 Production Field Joint Testing

8.1 Inspection Procedures

8.1-1 - The field joint surface preparation and pipe coating(s) applied in production shall be inspected and subjected to quality control (QC) testing in accordance with Table 4 and Table 5, and the Supplier- and Owner-accepted production field joint inspection and test plan (ITP).

8.1-2 - Unless and otherwise agreed by Owner not to include production testing, the production tests included in Table 5 shall be performed using simulated field joints prepared from pipe pups inserted temporarily into the field joint coating and production line.

8.1-3 - Unless and otherwise agreed by Owner to reduce number of tests included in Table 5, The inspections described in Table 5 shall be performed for the actual production field joints.

8.2 Pipe Pup Preparation

8.2-1 - Coated project pipe identical to the pipe being used in production shall be cut into pups approximately 2 feet (0.6 m) in length by Sub-Supplier for the purpose of preparing coated field joint specimens for production quality control (QC) testing.

8.2-2 - Sub-Supplier shall prepare the pups by removing the pipe coatings from the center of each pup to produce a simulated field joint whose geometry and dimensions are identical to those of the production field joints.

8.3 Test Field Joint Preparation and Scheduling

8.3-1 - Unless and otherwise agreed by Owner not to include production testing, at the startup of each production run, and at least once a day during production, Sub-Supplier shall insert a pipe pup into the production line and apply the production field joint coating system to the simulated field joint using the same application equipment, procedures, personnel, and materials used for the production field joints.

8.3-2 - If production is taking place in two or more shifts, then the preparation of test field joints shall be scheduled so that the number prepared is approximately the same for each shift.

8.3-3 - Supplier and Owner inspectors shall have the right to request that additional test field joints be prepared at any time if, in the opinion of the inspectors, there is reason to be concerned about the application, inspection, or quality of the field joint coating.

8.4 Special Considerations for Operations Offshore

8.4-1 - If the field joint coatings are being applied on a pipelay vessel offshore, then—subject to the acceptance of Supplier and Owner—the dimensions and configuration of the pipe pups and test field joints may be modified by Sub-Supplier to facilitate their handling, preparation, and inspection offshore.

8.4-2 - The redesigned test field joints shall be sufficiently similar to the actual field joints that any application and quality control (QC) problems with the actual field joints will be reproduced in the test field joints.

8.4-3 - Subject to Supplier and Owner inspector acceptance, the preparation of test field joints may be scheduled to minimize disruption of Sub-Supplier's offshore construction activities.

8.5 Test Specimen Preparation and Sectioning

8.5-1 - After application of the field joint coating, the pipe pup and simulated field joint shall be removed from the production line and allowed to cool and cure.

8.5-2 - After field joint has sufficient time to fully cool and cure (not to exceed one day), it shall be sectioned to permit Sub-Supplier and Owner's inspector(s) to inspect the interior of the insulated field joint and obtain specimens needed for the production tests specified in Table 4.

9 Field Joint Application

9.1 Pipe Cleaning

9.1-1 - Oily contamination clean up shall meet the following requirements:

9.1-1.1 - Exposed pipe surface at the field joint shall be inspected by Sub-Supplier for contamination before coating application.

9.1-1.2 - Any oily contamination on the pipe surface shall be removed using any of the following:

1. A procedure consistent with SSPC-SP 1.
2. Solvent cleaning.
3. A detergent wash.
4. Freshwater rinse.
5. A combination of these methods.

9.1-2 - Chloride contamination shall be resolved as follows:

9.1-2.1 - Chloride levels shall be measured on the surface of the pipe using a salt contamination meter, which uses conductivity measurements performed on a filter paper disk.

9.1-2.2 - If an unacceptable level of chloride contamination is found, the chlorides shall be removed using a high-

pressure rinse with deionized, filtered water.

9.1-3 - Pre-cleaning shall be done as follows:

1. High-pressure air may be used to remove soil, dust, and other loose materials from the pipe surface.
2. Abrasive sanding (flapper disc) or a wire brush may be used to remove material that is adhering to the pipe surface and cannot be effectively removed by other methods.
3. High pressure air used for pre-cleaning or other activities shall be tested per ASTM D4285.

9.1-4 - Pipe coating at the ends of the field joint shall be cleaned and stripped by scraping and/or sanding (flapper disc or similar), or scraping and sanding.

9.1-5 - Other appropriate methods shall be used to remove any contaminated or ultraviolet (UV) light damaged material and expose clean, fresh pipe coating to facilitate bonding between pipe and field joint coatings.

9.2 Blast Cleaning and Abrasive Monitoring

9.2.1 Blast Depth

9.2.1-1 - The exposed outer surface of the pipe at the field joint shall be grit blasted by Sub-Supplier to prepare the pipe surface and provide an anchor pattern for application of the fusion-bonded epoxy (FBE) coating.

9.2.1-2 - The depth of the blast profile shall be in the range of 50 to 100 microns (2 to 4 mils).

9.2.1-3 - Blast profile shall have been qualified during pre-qualification test (PQT) and demonstrated during pre-production test (PPT).

9.2.2 Blast Procedure

9.2.2-1 - The pipe surface blast shall be performed using a manual or automated high-pressure air blast system with angular steel or chilled iron grit.

9.2.2-2 - There shall be sufficient angular abrasive in the working mix to ensure that the required blast pattern is achieved at all times.

9.2.2-3 - This shall be achieved by the use of only new media or the continuous removal of dust and fines and frequent addition of new fresh grit in the blast station.

9.2.2-4 - Sub-Supplier's blast procedure shall specify the type of grit to be used and the addition frequency, as well as the grit monitoring and inspection procedures and schedules.

9.2.2-5 - Sub-Supplier shall provide justification for the blast parameters chosen.

9.2.2-6 - Sub-Supplier's blast procedure shall document and clearly identify the method and frequency of abrasive monitoring.

9.2.2-6.1 - At a minimum, there shall be one microscopic examination and one test per shift for conductivity of the

abrasive extract as specified in Section 13 Appendix.

9.2.2-6.2 - For subsea service, the conductivity of the blast media, as measured in Section 13 Appendix, shall not exceed 50 $\mu\text{mhos/cm}$. For burial/underground service, the conductivity of the abrasive shall not exceed 100 $\mu\text{mhos/cm}$.

9.2.3 Blast Surface Condition

9.2.3-1 - The blast process shall ensure that all rust and other dry contaminants are effectively removed from the pipe surface after blasting.

9.2.3-2 - The blast process shall be limited to ensure that the pipe surface is not over-worked from excessive blasting.

***Note:** Excessive blasting can produce a surface that is contaminated with microscopic flakes of steel foil (peaks that have rolled over or broken off) that can show up as significant steel dust contamination in a microscopic examination of the production test specimens.*

9.2.4 Humidity and Cleanliness

9.2.4-1 - Sub-Supplier shall maintain the blasted pipe surface at a temperature at least 5°F (3°C) above the dewpoint at all times. ASTM E337 shall be used to verify compliance of humidity levels.

9.2.4-2 - Any pipe not coated within two hours after blasting shall be reblasted.

9.2.4-3 - If the relative humidity is greater than 85 percent, the pipe shall be reblasted after only one hour.

9.2.4-4 - Pipe with visible flash rust on its surface shall be reblasted before fusion-bonded epoxy (FBE) application.

9.3 Fusion-Bonded Epoxy Coating Application

9.3.1 Pipe Heating

9.3.1-1 - Heating of the pipe and application of fusion-bonded epoxy (FBE) coating, copolymer adhesive (if applicable), and field joint coating shall be performed by Sub-Supplier in one continuous operation.

9.3.1-2 - Exposed pipe at the field joint shall be heated for application of the fusion-bonded epoxy (FBE) coating, using electrical induction coil heaters, after the blast.

9.3.1-3 - Sub-Supplier shall provide Supplier with a detailed heating procedure, including every relevant equipment parameter and documented test result showing the pipe surface temperature and temperature distribution achieved and depth of penetration (soak) of the preheat.

9.3.1-4 - These tests shall be witnessed by Supplier and Owner inspectors, and test procedures and results achieved shall be subject to Supplier and Owner review and acceptance.

9.3.2 Temperature Range

9.3.2-1 - The pipe temperature range used by Sub-Supplier for application of the fusion-bonded epoxy (FBE) pipe coating shall generally be in accordance with the manufacturer's recommendations, provided the minimum application temperature used is sufficient to ensure proper wetting of the pipe surface blast profile by the FBE.

9.3.2-2 - In some applications, particularly fusion-bonded epoxy (FBE) coatings intended for high service temperatures, it may be necessary for Sub-Supplier to tighten the application temperature range and increase minimum application temperature recommended by the FBE manufacturer to achieve adequate adhesion between pipe surface and FBE.

9.3.2-3 - Fusion-bonded epoxy (FBE) application temperature range and cure time used by Sub-Supplier shall be subject to the review and acceptance of Supplier and Owner.

9.3.2-4 - To protect against damage to the pipe steel, the pipe temperature shall not, at any time, be allowed to exceed 500°F (260°C). Any pipe that exceeds 500°F (260°C) and/or demonstrates discoloration (e.g., bluing) as a result of preheating shall be rejected.

9.3.2-5 - Sub-Supplier shall monitor the field joint preheat temperature used for application of fusion-bonded epoxy (FBE) pipe coating by using a calibrated digital thermometer and thermocouple contact probe that are acceptable to Supplier and Owner.

9.3.2-6 - The probe's condition shall be monitored frequently because there can be significant wear after repeated measurements on the blasted pipe surface.

9.3.2-7 - Sub-Supplier may also be required to use an optical pyrometer to measure pipe surface temperature during preheat to facilitate inspection and monitoring by Supplier's and Owner's inspector(s).

9.3.2-8 - Sub-Supplier shall check the optical pyrometer reading at least once per shift and at any other time requested by Supplier, Owner inspector(s), or both Supplier and Owner inspector(s), using a calibrated digital thermometer and thermocouple contact probe that are acceptable to Supplier and Owner.

9.3.3 Application

9.3.3-1 - The fusion-bonded epoxy (FBE) coating shall be applied to the heated pipe surface using automated powder spray equipment.

9.3.3-2 - The powder flow rate, application time, and movement and positioning of the spray heads shall be controlled to ensure the even distribution and correct thickness of the fusion-bonded epoxy (FBE).

9.3.3-3 - The fusion-bonded epoxy (FBE) application system shall be fully enclosed and include vacuum recovery to capture powder not adhering to the pipe surface.

9.3.3-4 - There shall be no recycling of the recovered fusion-bonded epoxy (FBE) powder.

9.3.3-5 - The thickness of the cured fusion-bonded epoxy (FBE) anticorrosion coating shall be in the range of 10 mils

to 20 mils (250 microns to 500 microns).

9.4 Process Variable Measurement

9.4-1 - Sub-Supplier shall monitor and record critical application parameters and process variables including the following, where applicable:

1. Fusion-bonded epoxy (FBE) application temperature.
2. Pipe coating preheat surface temperature and material softening.
3. Coating material extruder and injection temperature and pressure (may either be polypropylene or polyethylene).
4. Material mix ratio (may be polyurethane).
5. Ambient conditions.

9.4-2 - These measurement records shall be maintained at Sub-Supplier's coating facility or job site and made available to Supplier and Owner inspectors upon request for review and inspection.

9.4-3 - Summaries of this data, including maximum and minimum recorded values of each parameter for each shift, shall be included by Sub-Supplier in the weekly production reports provided to Supplier and Owner.

9.4-4 - All project records shall be maintained by Sub-Supplier for five years from termination of the contract. Owner may request certain project records to become a permanent part of final project documentation and included in the project databook.

9.5 Copolymer Adhesive Application (If Applicable)

9.5-1 - Sub-Supplier shall apply the copolymer adhesive onto the fusion-bonded epoxy (FBE)-coated pipe surface within the gel time of the FBE powder at the actual application pipe temperature.

9.5-2 - The copolymer adhesive shall be applied according to Sub-Supplier's accepted application method either in the form of a powder or in a tape wrap form.

9.5-3 - For application in the form of a powder, the copolymer adhesive shall be sprayed onto the pipe surface using the same automated system used to apply the fusion-bonded epoxy (FBE).

9.5-3.1 - The copolymer adhesive can be mixed with the fusion-bonded epoxy (FBE) powder during the transition from FBE to adhesive.

9.5-4 - Copolymer adhesive applied in a tape wrap form shall be applied immediately before the external anticorrosion coating with the same automated system.

9.5-5 - The average and minimum allowable thicknesses of the copolymer adhesive layer shall be as specified by Sub-Supplier and agreed to by Supplier and Owner.

9.6 Preparations for Field-Joint Coating Application

9.6-1 - Before application of the field joint coating, the pipe parent coating adjacent to the field joint location shall be cleaned by Sub-Supplier by air jetting, scraping, or jetting and scraping to remove contamination resulting from application of the fusion-bonded epoxy (FBE) and adhesive (if applicable).

9.6-2 - Any copolymer adhesive covering the fusion-bonded epoxy (FBE) shall be removed before FBE application to ensure a proper bond is established between the pipe parent FBE coating and the field joint FBE coating.

9.6-3 - Depending upon the type of field joint material used, the surface of the parent pipe coating adjacent to the field joint may be heated to promote adhesion between the parent pipe coating and field joint coating.

9.6-4 - If required, this heating shall be performed by using either a gas burner, radiant infrared, or hot air heater, and shall be sufficient to soften the pipe coating without oxidizing or degrading the coating surface.

9.6-5 - The temperature and softening of pipe overlaps shall be continuously monitored to ensure that the preheating is evenly distributed around the circumference of the pipe and applied equally at both ends of the field joint.

9.7 Polypropylene and Polyethylene Tape Wrap Coating

9.7-1 - Polypropylene (PP)- and polyethylene (PE)-based tape wrap coatings shall be applied by Sub-Supplier over the fusion-bonded epoxy (FBE) using a continuous, automated side wrap extrusion process.

9.7-2 - During this process, a polypropylene (PP)- or polyethylene (PE)-based copolymer adhesive shall be applied directly over the fusion-bonded epoxy (FBE) at the field joint and then overlain by a second layer of the PP or PE outer jacket material.

9.7-3 - Both layers shall be applied as a series of overlapping tape wraps composed of semi-molten material that cools and solidifies in place on the pipe.

9.7-4 - The copolymer adhesive shall be applied immediately after application of the fusion-bonded epoxy (FBE), before the FBE has completely gelled, to ensure adhesion to the FBE.

9.7.1 Injection Molded Polypropylene Coatings

9.7.1-1 - Injection molded polypropylene (IMPP) field joint coatings shall be applied by Sub-Supplier as a single layer that completely fills the gap between the parent pipe coating on either side of the field joint.

9.7.1-2 - Injection molded polypropylene (IMPP) coatings shall be applied by using high pressure to inject the molten polypropylene (PP) coating material into a reusable, rigid mold that surrounds the pipe at the field joint and overlaps pipe coating on either side of the field joint.

9.7.1-3 - The molten polypropylene (PP) shall be applied directly over the fusion-bonded epoxy (FBE) and a layer of copolymer adhesive spray shall be applied over the FBE.

9.7.2 Molded Polyurethane Field Joint Coatings

9.7.2-1 - Polyurethane-based field joint coatings shall be applied by Sub-Supplier as a single layer that completely fills the gap between the parent pipe coating on either side of the field joint by pouring or injecting the freshly mixed, uncured polyurethane material into a reusable, rigid mold that surrounds the pipe at the field joint and overlaps the pipe coating on either side of the field joint.

9.7.2-2 - The field joint coating shall be applied directly over the fusion-bonded epoxy (FBE) or adhesive (if applicable) and pre-heated pipe coating.

9.7.3 Flame-Sprayed Polypropylene or Polyethylene

9.7.3-1 - Flame-sprayed polyurethane and polyethylene (PE) coatings shall be applied by Sub-Supplier as a single, continuous layer that completely covers the field joint and overlaps the parent pipe coating on either side of the field joint.

9.7.3-2 - Flame-sprayed coatings shall be applied by feeding powdered polypropylene (PP) or polyethylene (PE) coating material that incorporates a copolymer adhesive through an oxygen rich flame, which liquefies the materials and allows them to be sprayed onto the surface of the field joint.

9.7.3-3 - The flame sprayed material shall be applied directly over the fusion-bonded epoxy (FBE) before the FBE has fully cured to ensure adhesion between the flame sprayed polypropylene (PP) or polyethylene (PE) and the FBE.

***Note:** It may be necessary to preheat the surface of the polypropylene (PP) or polyethylene (PE) pipe coating to ensure fusion between the two materials at the ends of the field joint.*

9.8 Shrink Sleeves

9.8-1 - Polypropylene (PP) and polyethylene (PE) shrink sleeves shall consist of a rectangular sheet of highly cross-linked PP or polyethylene that has a layer of polypropylene- or polyethylene-based adhesive applied on one side (the underside of the sleeve). If accepted by Supplier and Owner, polypropylene and polyethylene shrink sleeves shall be applied by Sub-Supplier over the field joint as a single piece that wraps completely around the pipe at the field joint and overlaps itself and the parent pipe coating at each end of the field joint.

9.8-2 - After being wrapped around the pipe, the outer surface of the shrink sleeve shall be heated with a torch to melt the underlying adhesive and shrink the cross-linked sleeve to form a tight fit around the pipe.

9.8-3 - A separate strip of the shrink sleeve material shall normally be applied over the longitudinal seam, where the shrink sleeve overlaps itself, to reinforce and help seal the seam.

9.8-4 - Shrink sleeves shall be applied either directly over the fusion-bonded epoxy (FBE) and adhesive before the FBE has fully cured or over the cured liquid anti-corrosion epoxy primer to ensure adhesion between the copolymer adhesive and the anti-corrosion primer.

9.8-5 - Some preheating of the pipe surface and surface of the pipe coating on either side of the field joint may be required to fully melt the adhesive.

9.8-6 - Shrink sleeves shall NOT be used on pipe that will be reeled or will pass through a pipe tensioner or be supported by heavily loaded track-type supports.

9.9 Field Joint Geometry and Overlap

9.9-1 - For coatings in which the outer jacket material is applied as a single layer wrapped or sprayed onto the pipe, the minimum thickness of the outer jacket material shall be 0.125 inch (3 mm).

9.9-2 - For coatings that are cast or injection molded around the pipe at the field joint, the thickness of the field joint material shall be a minimum of 0.125 inch (3 mm) greater than the outside diameter of the parent pipe coating, which will allow the field joint coating to completely fill the gap between the parent pipe coating and the mold.

9.9-3 - In all cases, field joint coating shall overlap the parent pipe coating by a minimum of 2 inches (50 mm) at each end of the field joint.

9.10 Rejectable Defects

9.10-1 - The surface of the cast or molded field joint coatings (e.g., injection molded polypropylene [IMPP], Polyurethane) shall be uniform in color.

9.10-2 - There shall be no discolored surface areas or swirls that can indicate incomplete curing or fusion of coating materials, improper mixing of coloring or foaming agents, or presence of degraded (e.g., overheated) materials.

9.10-3 - Applied field joint coating shall be free of holidays and internal voids, bubbles, or pores larger than 0.04 inch (1 mm) in diameter as specified in [Table 5](#).

9.10-4 - The surface of the coating shall be free of holes, tears, depressions, and blisters.

9.10-5 - There shall be no visible uncured or incompletely cured material.

9.10-6 - There shall be no gaps, separations, or other evidence of disbondment between the pipe and field joint coatings.

9.11 Application Procedure and Repair

9.11-1 - Every field joint repair procedure, materials, and equipment shall be fully documented by Sub-Supplier. Each repair procedure shall clearly specify the types of coating defects and maximum dimensions of defects that can be repaired using each procedure.

9.11-2 - All coating repair procedures, materials, and equipment shall be qualified by Sub-Supplier as part of the coating pre-qualification test (PQT).

9.11-3 - Any repair procedure(s) proposed by Sub-Supplier shall be documented, and a qualification test procedure shall be submitted to Supplier for review and acceptance at least thirty days before the pre-qualification test (PQT).

9.11-4 - The qualification test procedure may include the following tests:

1. Peel tests.
2. Tensile tests.
3. Cathodic disbondment and hot soak tests.
4. Permeability tests.
5. Full-scale reeling trials.
6. Other tests.

9.11-5 - All repair procedure qualification activities shall be witnessed by Supplier's inspectors.

9.11-6 - All coating repairs shall be witnessed and subject to acceptance by Supplier's inspectors. Sub-Supplier shall give sufficient notice to allow Supplier's inspectors to witness all coating repair activities.

9.12 Field Joint Geometry

9.12.1 Coating Cutback

9.12.1-1 - The geometry of the field joint system shall be proposed by Sub-Supplier based on configuration proposed and preferred application method.

9.12.1-2 - Unless a project specifies required cutback length, the preliminary value for the minimum cutback length shall be 15 inches (381 mm).

9.12.1-3 - The final production dimensions and tolerances for the coating cutback shall be established by Supplier after consultation with Sub-Supplier and the project's pipelay and automatic ultrasonic testing (AUT) Supplier.

9.12.2 Exposed Fusion-Bonded Epoxy Ring

9.12.2-1 - The cutback provided by the Sub-Supplier shall leave a clean, continuous exposed ring of the pipe's fusion-bonded epoxy (FBE) coating at least 0.4 inch (10 mm) in length at the toe of the cutback to allow continuity and an overlapping bond between the pipe's factory FBE coating and the FBE coating applied by Sub-Supplier.

9.12.2-2 - This exposed fusion-bonded epoxy (FBE) coating shall be preserved through the cleaning and blast procedures performed by Sub-Supplier before application of the field joint coatings.

10 Production Field Joint Quality Control Inspection, Testing, and Repairs

10.1 Quality Control and Inspection Procedures

10.1-1 - Sub-Supplier shall have written inspection procedures and suitable forms in place to do the following:

1. Monitor critical aspects of the surface preparation and field joint coating application processes.
2. Ensure compliance with acceptance criteria, the quality plan, inspection and test plan (ITP), and procedures established for the coating application.

10.1-2 - These plans and procedures shall be included in Sub-Supplier's quality control (QC) documentation provided to Supplier.

10.2 Production Testing – Test Field Joints

10.2-1 - The production tests performed on simulated field joints prepared using pipe pups shall be conducted on site by Sub-Supplier, as specified in Table 4, to monitor quality of the production field joint coatings and proper functioning of application equipment.

10.2-2 - The tests shall be performed at the beginning of each shift, or as deemed necessary or agreed upon by Supplier and Owner inspectors.

10.2-3 - Supplier's inspectors shall have the right to request additional tests to be performed in between regularly scheduled tests if the inspectors have any immediate concerns regarding the application or quality of field joint coatings.

10.2-4 - Sub-Supplier's field joint coating system shall successfully pass every production test listed in Table 5 for each production field joint and test field joint (pipe pup) prepared during production to be accepted by Supplier and Owner for use on the project.

10.3 Production Test Failures

10.3-1 - If any production field joint or test field joint fails any production test in Table 5, or any hidden defects or material failures are discovered, then the following actions shall be taken:

10.3-1.1 - The application process, equipment malfunction, or material defect that caused the failure shall be corrected immediately by Sub-Supplier.

10.3-1.2 - The failed test(s) shall be repeated until a successful outcome is achieved using newly prepared field joints.

10.4 Processing of Failed Field Joints

10.4.1 Defect Evaluation

10.4.1-1 - All test failures and coating defects found shall be evaluated by Sub-Supplier, Supplier, and Owner inspectors to determine the following outcomes:

10.4.1-1.1 - Field joint coating can be successfully repaired by Sub-Supplier.

10.4.1-1.2 - Observed defects will not adversely affect the coating performance and the affected field joints can be accepted by Supplier and Owner.

10.4.1-1.3 - Field joint coating shall be rejected.

10.4.1-1.4 - Field joint coatings rejected by Supplier, Owner, or Sub-Supplier shall be fully removed from the pipe.

10.4.1-2 - If any test failures or defects are noted in actual or test field joints prepared during production, in an onshore or offline production facility such as a spool base or double joint assembly line, every field joint prepared after the last successfully completed test before the failed test shall be identified and segregated to enable any required follow-up tests and repairs to be performed.

10.4.1-2.1 - The field joints affected shall be recoated, in accordance with Sub-Supplier's written stripping and recoating procedures accepted by Supplier and Owner.

10.4.2 Test Failure Documentation

10.4.2-1 - Sub-Supplier shall promptly report each test failure and identify every segregated field joint and the reason(s) for their rejection to Supplier and Owner inspectors.

10.4.2-2 - Sub-Supplier shall include this information in Sub-Supplier's weekly and final production reports to Supplier and Owner.

10.4.2-3 - Sub-Supplier shall prepare a detailed nonconformance report (NCR) documenting the following for each field joint and the measures put in place to prevent a reoccurrence:

1. Each test failure and cause of failure.
2. Field joints involved.
3. Steps taken to correct the failure (repair, stripping and recoating, etc.)

10.4.3 Repairs

10.4.3-1 - Each coating repair performed by Sub-Supplier shall be prepared in accordance with a qualified and accepted repair procedure, witnessed by Supplier and Owner inspectors, and subject to Supplier and Owner acceptance.

10.4.3-2 - Each repair shall be photographed and documented by Sub-Supplier.

10.4.3-2.1 - The size, location, and nature of the repair shall be reported to Supplier and Owner inspectors on at least a daily basis.

10.4.3-2.2 - The information about the repair shall be included in Sub-Supplier's weekly and final production reports provided to Supplier and Owner.

10.5 Coating Thickness Measurements

10.5.1 Magnetic Gauge

10.5.1-1 - Thickness measurements for fusion-bonded epoxy (FBE) adhesive and outer jacket coatings shall be made using a calibrated magnetic gauge acceptable to Supplier and Owner.

10.5.1-2 - The thickness of each coating shall be tested at multiple points around the pipe circumference to verify that coating thicknesses and uniformity are within allowable limits.

10.5.2 Holiday Testing

10.5.2-1 - Holiday testing on simulated test field joints shall be performed only on the fusion-bonded epoxy (FBE) at a minimum test voltage of 25kV in accordance with NACE SP0274.

11 Technical Requirements for Field Joint Coating System

11.1 Anticorrosion Coating Pre-Qualification Test Series

11.1-1 - All tests shall be performed on two separate samples taken from two different points on each of the three simulated field joints coated during the pre-qualification test (PQT).

Note: *Table 1 summarizes the minimum technical requirements for inspection and testing of the field joint anticorrosion coating system.*

Table 1: Field Joint Anticorrosion Coating Pre-Qualification and Pre-Production Test Series

Test	Procedure	Acceptance Criterion
Fusion-Bonded Epoxy (FBE) Only:		
Blast Preparation: Roughness Surface Finish Residual Dust Soluble Salt Contamination	ASTM D4417 Method C (Testex SM Tape) or ISO 8503-5 SSPC VIS 1 or ISO 8501-1 and ISO 8501-3 ISO 8502-3 additional tests (TBD) ISO 8502-6 or ISO 8502-9	2 to 4 mils (50 – 100 µm) SSPC-SP 10 or SA 2-1/2 Rating 1-2 for size and quantity Equal or lower than 20 mg/m2
Application Pipe Temperature	Optical pyrometer confirmed by calibrated thermocouple thermometer	Manufacturer's data sheet - Subject to Owner acceptance (typ. 450°F to 488°F [232°C to 253°C] for high-temperature FBE)
FBE Thickness	Magnetic or digital thickness gauge	10 mils (250 µm) minimum 20 mils (500 µm) maximum
Rectangle (Knife) Cut Adhesion	Rating per NACE SP0394, Appendix J	#1
Microscopic Exam: Visible Contamination Foaming (Porosity)	NACE SP0394 Appendix K NACE SP0394 Appendix G	< 20% Rating 1-3
Flexibility in Bending	NACE SP0394, Appendix H Procedure A – Mandrel Bend	2.5% pipe diameter FBE Only
Differential Scanning Calorimeter (DSC)	NACE SP0394 Appendix D with nitrogen purge	T _{g2} > T _{design} + 8°F (5°C) Delta T _g < 8°F (5°C)
UV Resistance	DIN 30670	< 20% Reduction in tensile strength and elongation at break
Cathodic Disbondment	NACE SP0394 Appendix F 24 hours @ 149°F (65°C), 3.5V 28 days @ 149°F (65°C), 1.5V	0.2 in. (5 mm) max. 0.28 in. (7 mm) max.
Hot Water Soak, Full Immersion	NACE SP0394 Appendix J: 24 hours @ 203°F (95°C) (Remove PP to evaluate) 28 days @ 203°F (95°C) (Remove PP to evaluate)	No edge disbondment. Adhesion Rating #2 max. Edge disbondment < 0.125 in. (3 mm) measured using a 2 mil (0.05 mm) feeler gauge. Adhesion Rating #2 max.
Coating + Copolymer Adhesive (If Applicable):		
Visual Examination	100% visual examination	Uniform texture and color, no holes or blisters
Thickness	Visual and digital thickness gauge examination	Continuous layer of adhesive (approx. 8 mils to 16 mils [200 µm to 400 µm])

11.2 Heat Shrink Sleeve Pre-Qualification Test Program

11.2-1 - If heat shrink sleeve (HSS) is applied with fusion-bonded epoxy (FBE) and adhesive, it shall meet all relevant properties below.

Note: *Table 2 summarizes the minimum technical requirements for qualification, inspection, and testing of the heat shrink sleeve (HSS) field joint coating system when applied over liquid epoxy.*

Table 2: Field Joint Heat Shrink Sleeve (HSS) Pre-Qualification and Pre-Production Test Series

Property	Test Method	PE HSS Acceptance Criteria	PP HSS Acceptance Criteria
Liquid Epoxy Wet Film Thickness (WFT)	WFT Gauge	>10 mils (250 μ m)	
Visual Inspection of Applied HSS Coating System	Visual	HSS fully conforms to the underlying field joint surface. Adhesive flows around outer circumference of HSS. A blemish that has no detrimental affect may be acceptable if it meets the other criteria of being smooth, bonded, and has no inclusions.	
Overlap onto Adjacent Factory Coating	Tape Measure	> 2 in. (50 mm)	
Holiday Detection @ 5 kV + 5 kV/mm	ISO 21809-1	No Holidays	
Thickness of Applied HSS Coating System	Thickness Gauge	> 99 mils (2.5 mm) on body > 74 mils (1.875 mm) on weld	
Adhesion to the Steel Surface at Ambient Temperature	ISO 21809-1 (10 mm/min)	> 381N/in. (150 N/cm)	>635 N/in. (250 N/cm)
Adhesion to the Factory Coating at Ambient Temperature	ISO 21809-1 (10 mm/min)	> 381 N/in. (150 N/cm)	>635 N/in. (250 N/cm)
Adhesion to the Steel Surface at Elevated Temperature	ISO 21809-1 (10 mm/min)	> 72 N/in. (30 N/cm) @ 176°F (80°C)	> 152 N/in. (60 N/cm) @ 230°F (110°C)
Adhesion to the Factory Coating at Elevated Temperature	ISO 21809-1 (10 mm/min)	> 72 N/in. (30 N/cm) @ 176°F (80°C)	> 152 N/in. (60 N/cm) @ 230°F (110°C)
Adhesive Lap Shear Strength at 73°F (23°C)	ISO 21809-3	>3225 N/in ² (> 500 N/cm ²)	
Adhesive Lap Shear Strength at Elevated Temperature	ISO 21809-3	> 645 N/in ² (> 100 N/cm ²) @ 176°F (80°C) for PE > 645 N/in ² (> 100 N/cm ²) @ 230°F (110°C) for PP	
Impact Resistance	ISO 21809-3	5 Nm/mm (5 J/mm) of coating thickness	
Resistance to Penetration at Elevated Temperature	DIN 30670	≤ 0.0158 in. (≤ 0.4 mm) of penetration @ 176°F (80°C)	≤ 0.0158 in. (≤ 0.4 mm) of penetration @ 230°F (110°C)
Resistance to Cathodic Disbondment, 28 days at 73°F (23°C)	ISO 21809-3	≤ 0.197 in. (≤ 5 mm)	
Resistance to Cathodic Disbondment, 28 days at T _{max}	ISO 21809-3	≤ 0.56 in. (≤ 15 mm)	
Hardness	ISO 868 or ASTM D2240	55 Shore D	65 Shore D

11.3 Coating System Pre-Qualification and Pre-Production Test Series

11.3-1 - Tests shall be performed using material samples taken from three different pipes in the as-produced condition, plus one sample each from the Bending Test and Simulated Service Test pipes (the * asterisk refers to items in [Table 3](#)).

11.3-2 - These tests shall also be conducted on one sample each taken from the Reeling Trial and Simulated Service Test pipes. If the outer jacket material (e.g., polypropylene [PP] or polyurethane) is bonded to the anticorrosion coating, a flapper wheel, grinder, or machining shall be used to reduce the thickness of the jacket material to one millimeter or less after the test and before the evaluation (the ** asterisks refer to items in [Table 3](#)).

11.3-3 - The Hot Water Soak Test shall be performed for the complete anticorrosion coating system including the protective outer jacket. Disbondment shall be measured along the edges of the specimen at a distance of 0.6 inches (15 mm) from the corners of the specimen (the *** asterisks refer to items in [Table 3](#)).

Note: [Table 3](#) summarizes the minimum technical requirements for the inspection and testing of the complete field joint coating system.

Table 3: Full Coating System Pre-Qualification and Pre-Production Test Series

Test	Procedure/References	Acceptance Criterion
Tensile Strength at Yield*	ASTM D638	≥ 2610 psi (≥ 18MPa) @ 73.4°F (23°C)
Elongation at Break*	ASTM D638	> 400% Average of 5 values, > 200% Worst Case, all @ 73.4°F (23°C)
Water Absorption*	ASTM D570	< 3% @ 7 days, 145°F (65°C)
Hardness*	ASTM D2240	Per approved design
Density*	ASTM D792	Per approved design
Thickness*	Caliper or machinist rule	Total thickness: +/-10% of design value at any point
Abrasion Resistance	ASTM D4060	< 0.0007 oz. (20 mg) /1,000 cycles
field joint-to-Pipe-Coating Interface Adhesion*	ASTM D638	> 750 psi (5.1 MPa) @ 73.4°F (23°C)
Porosity*	Caliper or Microscope	Isolated; maximum diameter < 0.06 in. (1.5 mm)
Impact Resistance	Sub-Supplier Proposal / Owner Acceptance	Sub-Supplier Proposal / Owner Acceptance
Cathodic Disbondment**	NACE SP0394 Appendix F 24 hours @ 149°F (65°C), 3.5V (Remove PP to evaluate) 28 days @ 149°F (65°C), 1.5V (Remove PP to evaluate)	0.2 in. (5 mm) max. 0.28 in. (7 mm) max.
Hot Water Soak, Full Immersion **, ***	NACE SP0394 Appendix J: 24 hours @ 203°F (95°C) (Remove PP to evaluate) 28 days @ 203°F (95°C) (Remove PP to evaluate)	No edge disbondment. Adhesion Rating #2 max. Edge disbondment < 0.125 in. (3 mm) measured using a 2-mils (0.05-mm) feeler gauge. Adhesion Rating #2 max.

11.4 Additional Full-Scale Pre-Qualification Test Series

11.4-1 - With Supplier and Owner review and acceptance, some tests may be combined with tests performed for the factory-applied pipe coating by using shared test specimens.

11.4-2 - Applicability of the tests shall be determined on the basis of project requirements and the pipelay installation method selected (see *asterisk items in Table 4).

11.4-3 - The following shall apply for this test (see **asterisks items in Table 4):

11.4-3.1 - There shall be no degradation of pipe coatings or materials as a result of hydrolysis or test conditions.

11.4-3.2 - There shall be no internal or external cracking, disbondment, or tearing of pipe coating materials or any separation between pipe and field joint coating materials.

11.4-3.3 - There shall be no significant internal voids as a result of material failure, entrapped gases, or failure to fully fill the mold, die, or form during manufacture.

11.4-3.4 - There shall be no weakened or unbonded internal interfaces or internal defects such as materials that have been improperly mixed, contaminated, oxidized, or otherwise degraded or improperly prepared.

11.4-3.5 - Water shall not penetrate the pipe coatings and reach the pipe surface.

Note: Table 4 summarizes the minimum technical requirements for additional full-scale test series to be performed for the combined pipe and field joint coating systems.

Table 4: Full-Scale Combined Pre-Qualification Test Series

Test	Procedure/Reference	Acceptance Criterion
Full System on Pipe + Field Joint Sections		
Simulated Service Test *,**	Industry standard practice: Sub-Supplier proposal / Owner acceptance	Minimum 28 days @ Design temperature and pressure
Pipe Straightener and Equipment Handling Test*,**	Sub-Supplier proposal / Owner acceptance	Sub-Supplier proposal / Owner acceptance
VIV Suppression (Strakes) Shear Adhesion Test*	Sub-Supplier proposal / Owner acceptance	Sub-Supplier proposal / Owner acceptance
Reeling Trial*,**	Industry standard practice: 1. Bend radius not greater than minimum reel lay pipe radius. 2. Minimum of four bending + straightening cycles per pipe specimen. 3. Maximum coating temperature during test ≤ 20°F (−6°C)	Minimum of three consecutive successful tests per system for acceptance. (See item <u>1.4-3.3.**</u>)
Tensioner Pad Test*,**	Sub-Supplier proposal / Owner acceptance	Sub-Supplier proposal / Owner acceptance
Pull Tube Test*,**	S Sub-Supplier proposal / Owner acceptance	Sub-Supplier proposal / Owner acceptance

11.5 Production Quality Control Tests Based on Simulated Field Joints

11.5-1 - Unless otherwise agreed to by Owner to reduce the number of tests included in Table 5, tests shall be performed on a once-per-shift, once-per-day, or once-per-stalk basis.

11.5-2 - Tests shall be performed on material samples cut from a simulated field joint prepared using a temporary pipe pup that is inserted into Sub-Supplier's production line on a once-per-shift, once-per-day, or once-per-stalk basis, as specified in the scope of work (SOW).

11.5-3 - Sub-Supplier may propose alternative test procedure for Supplier and Owner acceptance when conditions (e.g., offshore application) or limited test facilities restrict Sub-Supplier's ability to perform required tests (see *asterisk items in Table 5).

11.5-4 - The Hot Water Soak Test shall be performed for the complete anticorrosion coating system including the protective outer jacket. Disbondment shall be measured along the edges of the specimen at a distance of 0.6 inches (15 mm) from the corners of the specimen (see **asterisks items in Table 5).

Note: Table 5 summarizes technical requirements for production tests to be performed for the field joint coatings using specimens derived from pipe pups (with simulated field joints) that are inserted into the production coating line.

Table 5: Production Quality Control Tests Based on Simulated Field Joints

Test	Procedure	Acceptance Criterion
Field Joint Corrosion Coating + Adhesive + Jacket:		
Blast Preparation: Roughness Surface/Standard Finish Residual Dust Soluble Salt Contamination	ASTM D4417 Method C (Testex Tape) or ISO 8503-9 SSPC VIS 1 or ISO 8501-1 and ISO 8501-3 ISO 8502-3 ISO 8502-6 or ISO 8502-9	2 to 4 mils (50 to 100 µm) SSPC-SP 10 or SA 2-1/2 Rating 1-2 for size and quantity Equal or lower than 20 mg/m2
Application Pipe Temperature	Optical pyrometer confirmed by calibrated thermocouple thermometer	Manufacturers data sheet – Subject to Owner acceptance (typ. 450°F to 488°F [232°C to 253°C] for high temperature FBE)
FBE Thickness	Magnetic or digital thickness gauge	mils (250 µm) minimum 20 mils (500 µm) maximum
Rectangle (Knife) Cut on FBE with Jacket Removed	Rating per NACE SP0394, Appendix J	#1
Microscopic Exam of FBE: Visible Contamination Foaming (Porosity)	NACE SP0394 Appendix K NACE SP0394 Appendix G	< 20% Rating 1-3
Flexibility in Bending*	NACE SP0394, Appendix H, Procedure A – Mandrel Bend at 32°F (0°C)	2.5°/ pipe diameter FBE only
DSC on FBE*	NACE SP0394 Appendix D with nitrogen purge	Tg2 > Tdesign + 8°F (5°C) Delta Tg < 8°F (5°C)
Porosity	Caliper or microscope	Isolated: Maximum diameter < 0.06 in. (1.5 mm)
Visual Examination on Cut Sections (PP, PE, polyurethane, shrink sleeve, etc.)	100% visual examination	Solid, smooth surface, only Isolated spherical voids < 0.06 in. (1.5 mm) diameter, no thru holes, cracks, tears, disbondment, gaps or blisters, no contaminated, or degraded material
field joint-to-Pipe-Coating Interface Adhesion*	ASTM D638	> 750 psi (5.1 MPa) @ 73.4°F (23°C)
Peel on 1- to 3-mm Thickness*	DIN 30670	> 26 lbs./in. @ 131°F (45 N/cm @ 55°C)
Cathodic Disbondment	NACE SP0394, Appendix F, 24 hours @ 149°F (65°C), 3.5V	< 0.2 in. (5 mm) maximum
Hot Water Soak**	NACE SP0394, Appendix J, 24 hours @ 203°F (95°C)	No edge disbondment, adhesion rating #2 max.
Thickness	Caliper or machinist's rule	Total thickness: +/-10% of design value at any point

11.6 Production Quality Control Tests for Field-Joint Coating

Note: *Table 6 summarizes the minimum technical requirements for production tests to be performed for every production field joint.*

Table 6: Production Quality Control Test Series for Applied Field-Joint Coating

Test	Procedure	Acceptance Criterion
Field Joint Coating		
Visual Examination	100 percent Visual Examination	Solid surface, uniform color, no depressions, holes, bubbles, blisters, tears or pores, no visible uncured or incompletely cured material, no gaps, separations, or other evidence of disbondment.

11.7 Tests to Verify Properties of Raw Materials

Note: *Table 7 summarizes technical requirements for quality control (QC) tests to be used for verifying the quality of raw materials used to produce the field joint coating.*

Table 7: Raw Material Properties Verification

Material	Property	Reference
Field Joint Coating Raw Material Testing:		
FBE [Note 1]	HR, Tg0, Cured Tg Gel Time	NACE SP0394 Appendix D NACE SP0394 Appendix C
Copolymer Adhesive [Note 1]	Softening point Melt flow index (MFI)	ISO 306 ISO 1133 or ASTM D1238
Solid Polypropylene / polyethylene / polyurethane*	Softening point Tensile strength and elongation MFI Density, impact (Charpy)	ISO 306 ASTM D638 ISO 1133 or ASTM D1238 ASTM D792 ISO 179-1
Note 1: Tests based on manufacturers' recommended values.		

12 Appendix—Additional References

The following references are not cited in this specification but are listed here for further reference.

American Petroleum Institute (API)

API Spec 5L Line Pipe

API RP 5L1 Recommended Practice for Railroad Transportation of Line Pipe

API RP 5LW Recommended Practice for Transportation of Line Pipe on Barges and Marine Vessels

API Spec Q1 Specification for Quality Management System Requirements for Manufacturing Organization for the Petroleum and Natural Gas Industry

ASTM International (ASTM)

ASTM C518 Standard Test Method for Steady-State Thermal Transmission Properties by Means of the Heat Flow Meter Apparatus

ASTM D695 Standard Test Method for Compressive Properties of Rigid Plastics

ASTM D1621 Standard Test Method for Compressive Properties of Rigid Cellular Plastics

ASTM D1622 Standard Test Method for Apparent Density of Rigid Cellular Plastics

ASTM D3748 Standard Practice for Evaluating High-Density Rigid Cellular Plastics

Canadian Standards Association (CSA)

CSA Z245.20 Plant-applied External Fusion Bond Epoxy Coating for Steel Pipe

CSA Z245.21 External Polyethylene Coating for Pipe

International Organization for Standardization (ISO)

ISO 9001 Quality Management Systems – Requirements

Association for Materials Protection and Performance (AMPP) (previously NACE International)

NACE RP287 Recommended Practice for Field Measurement of Surface Profile of Abrasive Blast Cleaned Steel Surface Using a Replica Tape

13 Appendix—Conductivity Monitoring of Abrasive Extracts

13.1 Scope

This method describes a procedure for monitoring the quality of the abrasive blast media in machines that recycle the abrasive. An increased conductivity level may indicate the introduction of an ionic contaminant into the abrasive from the pipe surface.

13.2 Materials

Materials for conductivity monitoring of abrasive extracts:

- 8-ounce (240-mL) bottle; glass or polypropylene (PP).
- 2-ounce (60-mL) bottle.
- 3.4-ounce (100-mL) graduated cylinder.
- Distilled or deionized water with a conductivity of less than 3 $\mu\text{mhos/cm}$.
- Conductivity meter (calibrated with appropriate liquid conductivity standard).

13.3 Procedure

Procedure for conductivity monitoring of abrasive extracts:

1. Measure a 3.4-ounce (100-mL) volume of abrasive into the 8-ounce (240-mL) bottle.
2. Add 3.4-ounce (100-mL) distilled or deionized water to the abrasive in the 8-ounce (240-mL) bottle.
3. Cap the bottle and shake vigorously for 60 seconds.
4. Decant the water extract into the 2-ounce (60-mL) bottle.
5. Measure and record the conductivity of the water with the conductivity meter.
6. Alternatively, titrate the water sample according to ASTM D512 to determine the chloride level in the water.
7. ASTM D4940 may also be used as test method for conductimetric analysis of water-soluble ionic contamination of blasting abrasives.